



NSBM Green University

Faculty of Computing

CN205.3 -Voice and Video over IP

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Course Work

Group L

Environment & VersionsHost & VM: VMware Workstation (VM for FreeBSD)

- Guest OS: FreeBSD 13.x (installed in VM)
- Asterisk: asterisk18 (pkg install used)
- Softphones used: Zoiper, Linphone (mobile and desktop)
- Notes: The FreeBSD package build used for this lab does not include PJSIP or TLS/SRTP support in its SIP module; this affects Lab 7 and Lab 8 testing.

Summary of Labs (1–9)

Lab	Objective	Result	Key Errors (if any)
Lab 1	FreeBSD installation & networking	VM online; network configured	No route to host (gateway missing)
Lab 2	Install Asterisk	Asterisk running (pkg)	portsnap mirrors error -> used pkg
Lab 3	Dialplan & hello-world	Playback works	Edited wrong path initially
Lab 4	Softphone registration	Zoiper register	408 Request Timeout, 401 Unauthorized
Lab 5	Extension-to-extension calls	Calls successful	404 Not Found until both registered
Lab 6	PF firewall	PF enabled; SIP/RTP allowed	ICMP blocked initially
Lab 7	TLS for SIP	Config prepared; TLS not available	No OpenSSL in chan_sip; pjsip missing
Lab 8	SRTP	Config prepared; SRTP not available	res_srtp.so missing
Lab 9	NAT & Security	Fail2ban + NAT documented	NAT complexity; easier with Bridged

Detailed Lab Reports

Lab 1 – FreeBSD VM Installation & Networking

Objective: Install FreeBSD guest and ensure network connectivity inside VMware.

High-level steps:

- Install FreeBSD in VMware Workstation.
- Set network adapter to NAT for initial internet access, or Bridged for direct LAN connectivity.
- Configure `/etc/rc.conf` with DHCP for `em0`.

Commands & configuration :

```
# Verify network interfaces
ifconfig em0

# Manually request DHCP (if needed)
dhclient em0

# Check route table
netstat -rn

# Add default gateway (if missing)
route add default 10.0.2.2
```

Expected output / behaviour:

```
# Expected if DHCP works

inet 192.168.220.129 netmask 0xffffffff broadcast 192.168.220.255

# Expected ping success

PING 8.8.8.8: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=20.0 ms
```

Observed output / errors:

```
# Observed errors in session

ping: sendto: No route to host

# ifconfig output sometimes blank for IPv4 in bridged mode
```

Root cause analysis (errors):

- DHCP assigned IP but no default route (missing gateway).
- Bridged mode sometimes failed due to host Wi-Fi adapter not allowing bridge DHCP.
- Fixes applied / Troubleshooting steps:
- Added default route manually: `route add default 10.0.2.2` (or appropriate gateway for your NAT).
- If Bridged fails, use NAT or set Bridged to the host Wi-Fi adapter explicitly.
- Restart netif: `service netif restart`; `service routing restart`

The screenshot shows a VMware Workstation interface with a single virtual machine named 'FreeBSD version 10 and earlier'. The main window displays the terminal output of the FreeBSD boot process. The boot sequence includes clearing the /tmp directory, starting sysinit, mounting file systems, starting cron, sendmail, and sshd. A login prompt is shown, and the user 'root' has successfully logged in. The terminal output is as follows:

```

Clearing /tmp (X related).
Starting sysinit.
No core dumps found.
Mounting late filesystems:.
Starting cron.
Starting sendmail_submit.
Starting sendmail_sendmail.
Performing sanity check on sshd configuration.
Starting sshd.
Configuring vt: blanktime.
Starting asterisk.
Starting background file system checks in 60 seconds.

Sat Sep 28 02:48:42 +0530 2025

FreeBSD/amd64 (hmemsiac) (tty0)

login: Sep 28 02:48:43 asterisk(h591): rc_read_config: can't open /etc/radinscli
ent-ng/radinscli.conf: No such file or directory

FreeBSD/amd64 (hmemsiac) (tty0)

login: root
Password:
root@: #
root@: #
root@: #
root@: #
root@: #
root@: #
root@: #
root@: #
root@: #
root@: #
root@: # ping 0.0.0.0
PING 0.0.0.0 (0.0.0.0): 56 data bytes
64 bytes from 0.0.0.0: icmp_seq=0 ttl=120 time=86.198 ms
64 bytes from 0.0.0.0: icmp_seq=1 ttl=120 time=72.225 ms
64 bytes from 0.0.0.0: icmp_seq=2 ttl=120 time=88.594 ms
64 bytes from 0.0.0.0: icmp_seq=3 ttl=120 time=125.267 ms
64 bytes from 0.0.0.0: icmp_seq=4 ttl=120 time=73.647 ms
64 bytes from 0.0.0.0: icmp_seq=5 ttl=120 time=76.898 ms
64 bytes from 0.0.0.0: icmp_seq=6 ttl=120 time=74.450 ms
64 bytes from 0.0.0.0: icmp_seq=7 ttl=120 time=67.581 ms
64 bytes from 0.0.0.0: icmp_seq=8 ttl=120 time=96.738 ms
^C
Suspended
root@: #

```


- Enable asterisk service in rc.conf and start the service.
- Open Asterisk CLI and confirm running status.

Commands & configuration :

```
# Update package repo (if network ok)

pkg update -f

pkg install asterisk18

# Enable/start asterisk

sysrc asterisk_enable=YES

service asterisk start

# Access console

asterisk -rvvv
```

Expected output / behaviour:

```
# Expected Asterisk to start and show banner

Asterisk 18.x.x, Copyright (C) ...

Asterisk Ready
```

Observed output / errors:

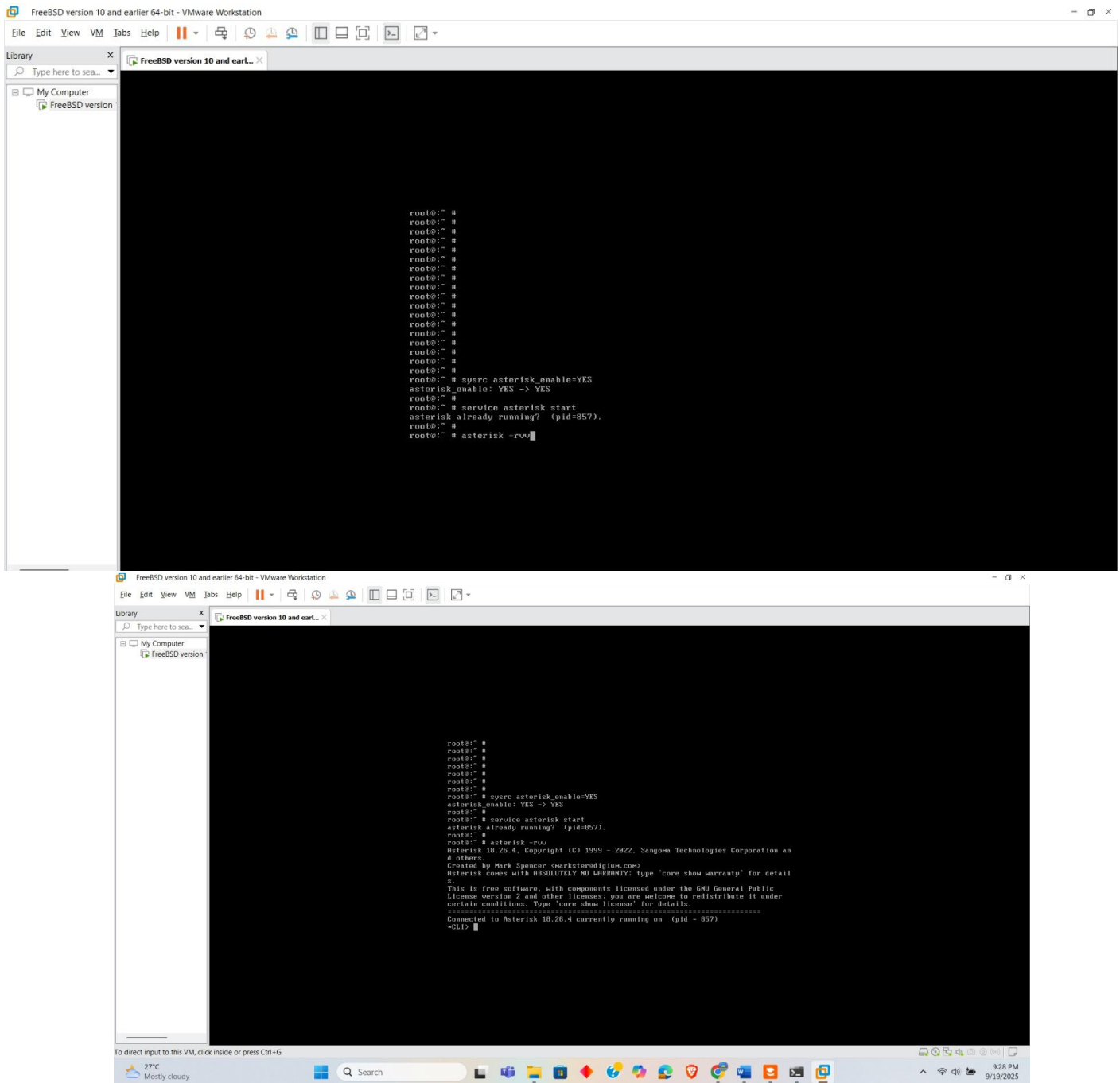
```
# Observed

portsnap fetch extract failed earlier due to no mirrors (network issue).
```

Root cause analysis (errors):

- portsnap fetch failed because internet/network was not configured correctly or DNS was blocked by PF earlier.
- Fixes applied / Troubleshooting steps:
- Used 'pkg install asterisk18' after resolving networking and disabling PF for initial package fetch.
- Notes: If pkg fails repeatedly, check /etc/resolv.conf and ensure DNS works (e.g., nameserver 8.8.8.8).

```
Pkg install succeeded later after network fixed.
```



High-level steps:

- Edit /usr/local/etc/asterisk/extensions.conf
- Add from-internal context with extension 100 playing hello-world
- Reload dialplan and test via console dial or softphone

Commands & configuration :

```
# File: /usr/local/etc/asterisk/extensions.conf
```

```
[from-internal]
```

```
exten => 100,1,Answer()
```

```
    same => n,Wait(1)
```

```
    same => n,Playback(hello-world)
```

```
    same => n,Hangup()
```

```
# After saving
```

```
asterisk -rx "dialplan reload"
```

```
# Test from CLI
```

```
console dial SIP/1001/100
```

Expected output / behaviour:

```
# Expected CLI logs
```

```
-- Executing [100@from-internal:1] Answer("SIP/1001-0000000", "0") in new stack
```

```
-- Playing 'hello-world' (language 'en')
```

Observed output / errors:

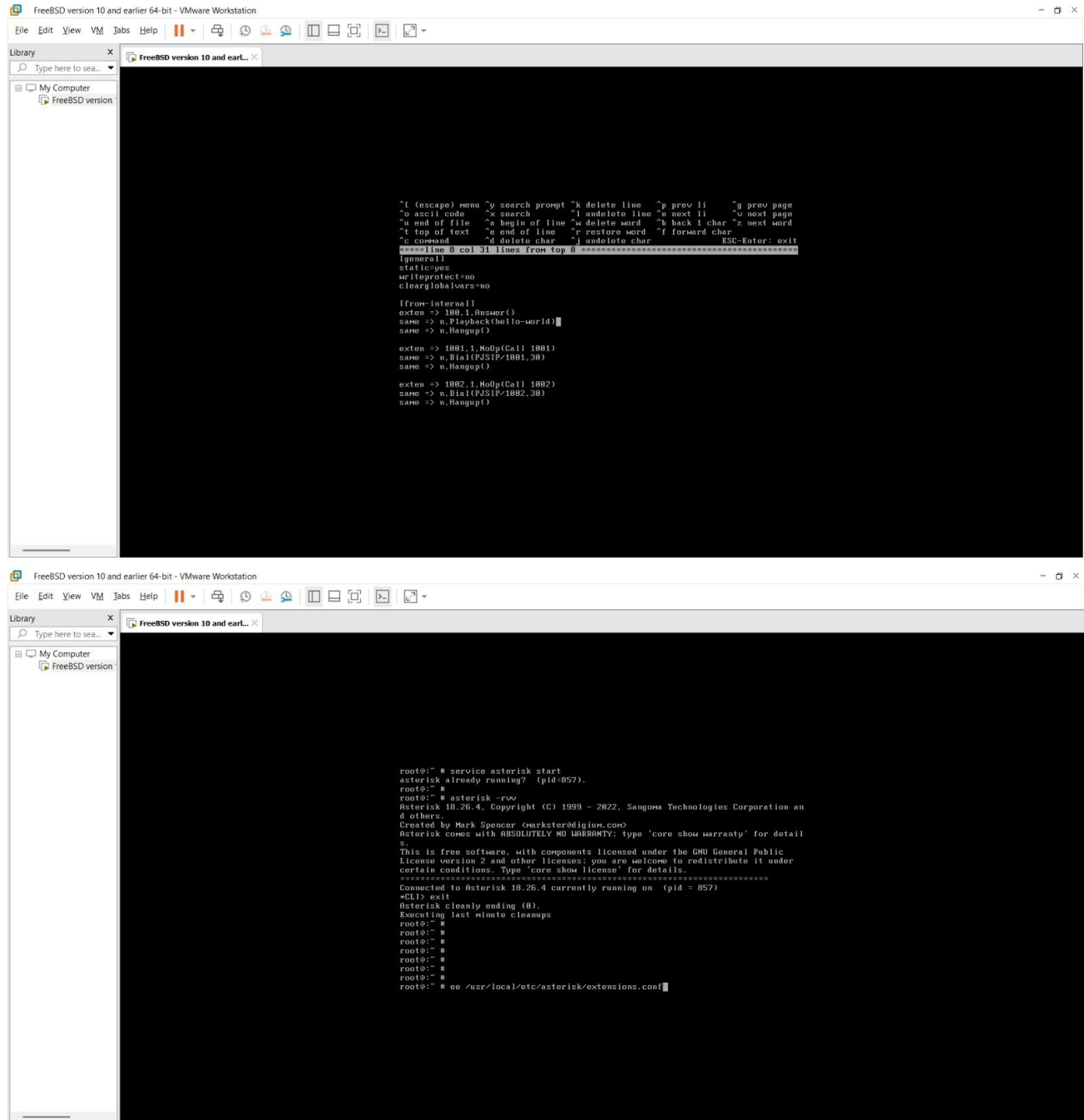
```
# Observed
```

```
Initially edited wrong directory (/etc/asterisk/extensions.conf) - no effect  
until corrected to /usr/local/etc/asterisk/extensions.conf
```

Root cause analysis (errors):

- Configuration file locations differ on FreeBSD: /usr/local/etc/asterisk/ is the real path for pkg installs.

- Fixes applied / Troubleshooting steps:
- Edit /usr/local/etc/asterisk/extensions.conf and reload dialplan with 'asterisk -rx "dialplan reload"'



Lab 4 – Softphone Registration (Zoiper / Linphone)

Objective: Register softphones to Asterisk and perform basic calls.

High-level steps:

- Create SIP accounts in sip.conf (or pjsip.conf if pjsip available).
- Configure Zoiper with domain/IP, username and password.
- Register and observe CLI for registration messages.

Commands & configuration :

```
# Example: chan_sip (sip.conf)

[general]

context=from-internal

bindport=5060

bindaddr=0.0.0.0

transport=udp

[1001]

type=friend

host=dynamic

secret=pass1001

context=from-internal

disallow=all

allow=ulaw
```

Expected output / behaviour:

```
# Expected in Asterisk CLI when Zoiper registers

Registered SIP '1001' at 192.168.1.50:5060

# Zoiper shows 'Registered'
```

Observed output / errors:

```
# Observed errors during session

- Zoiper: 408 Request Timeout (REGISTER sent but no response)
```

- Zoiper: 401 Unauthorized (wrong password)

Root cause analysis (errors):

- 408 Request Timeout: VM network (NAT) prevented direct packets from phone to VM or wrong port forward; Bridged mode was needed for mobile device registration.
- 401 Unauthorized: incorrect secret in sip.conf or Zoiper settings.
- Fixes applied / Troubleshooting steps:
- Switch VM network to Bridged (works when host and client are on same LAN) OR configure NAT port forwarding in VMware for UDP 5060 and RTP range.
- Verify user secret in /usr/local/etc/asterisk/sip.conf and password in Zoiper.

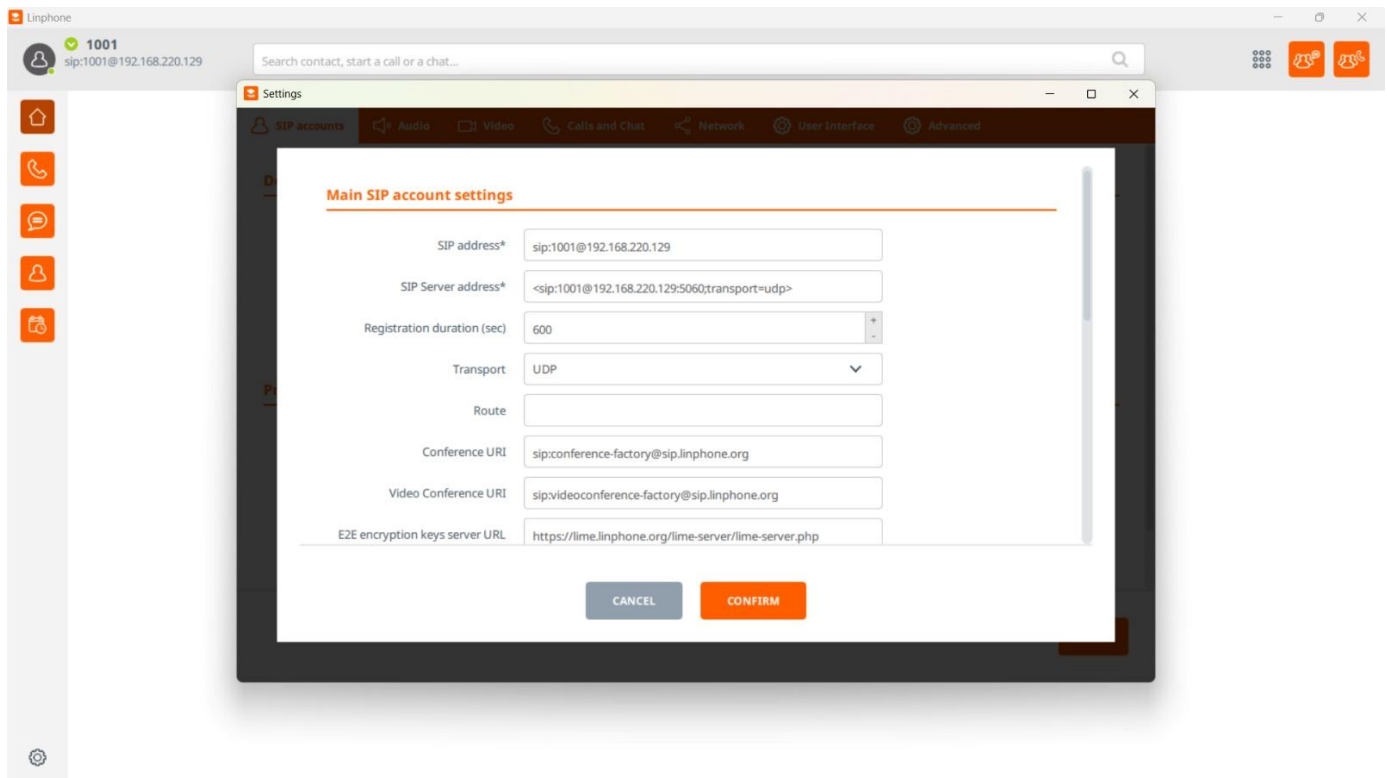
```
FreeBSD version 10 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
FreeBSD version

^f (escape) menu ^y search prompt ^k delete line ^p prev li ^g prev page
^o ascii code ^x search ^l undelate line ^n next li ^v next page
^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete char ^j undelate char ESC-Enter: exit
general 1 col 15 lines from top 1
[transport-udp]
type=transport
protocol=udp
bind=0.0.0.0:5060
[1001]
type=endpoint
context=from-internal
disallow=all
allow=law
auth=1001-auth
aors=1001
[1001-auth]
type=auth
auth_type=userpass
username=1001

FreeBSD version 10 and earlier 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
FreeBSD version

^f (escape) menu ^y search prompt ^k delete line ^p prev li ^g prev page
^o ascii code ^x search ^l undelate line ^n next li ^v next page
^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete char ^j undelate char ESC-Enter: exit
general 1 col 8 lines from top 1
[general]
static=yes
writeprotect=no
clearglobalvars=no
[from-internal]
exten => 1001.1,MoOp(Call 1001)
same => n,Bial(PJSIP/1001,30)
same => n,Hangup()
exten => 1002.1,MoOp(Call 1002)
same => n,Bial(PJSIP/1002,30)
same => n,Hangup()

file "/usr/local/etc/asterisk/extensions.conf", 13 lines
```



Lab 5 – Extension-to-Extension Calling

Objective: Call from 1001 to 1002 and confirm two-way audio.

High-level steps:

- Configure two users in sip.conf
- Configure dialplan in extensions.conf for direct SIP dialing
- Register both phones and call

Commands & configuration :

```
# sip.conf entries already included for 1001 and 1002

# extensions.conf entries

[from-internal]

exten => 1001,1,Dial(SIP/1001,30)
```

```
same => n, Hangup()

exten => 1002,1,Dial(SIP/1002,30)

same => n, Hangup()
```

Expected output / behaviour:

```
# Expected: call initiated and answered, audio RTP negotiated.

-- Executing [1002@from-internal:1] Dial("SIP/1001-0000001", "SIP/1002,30") in
new stack
```

Observed output / errors:

```
# Observed

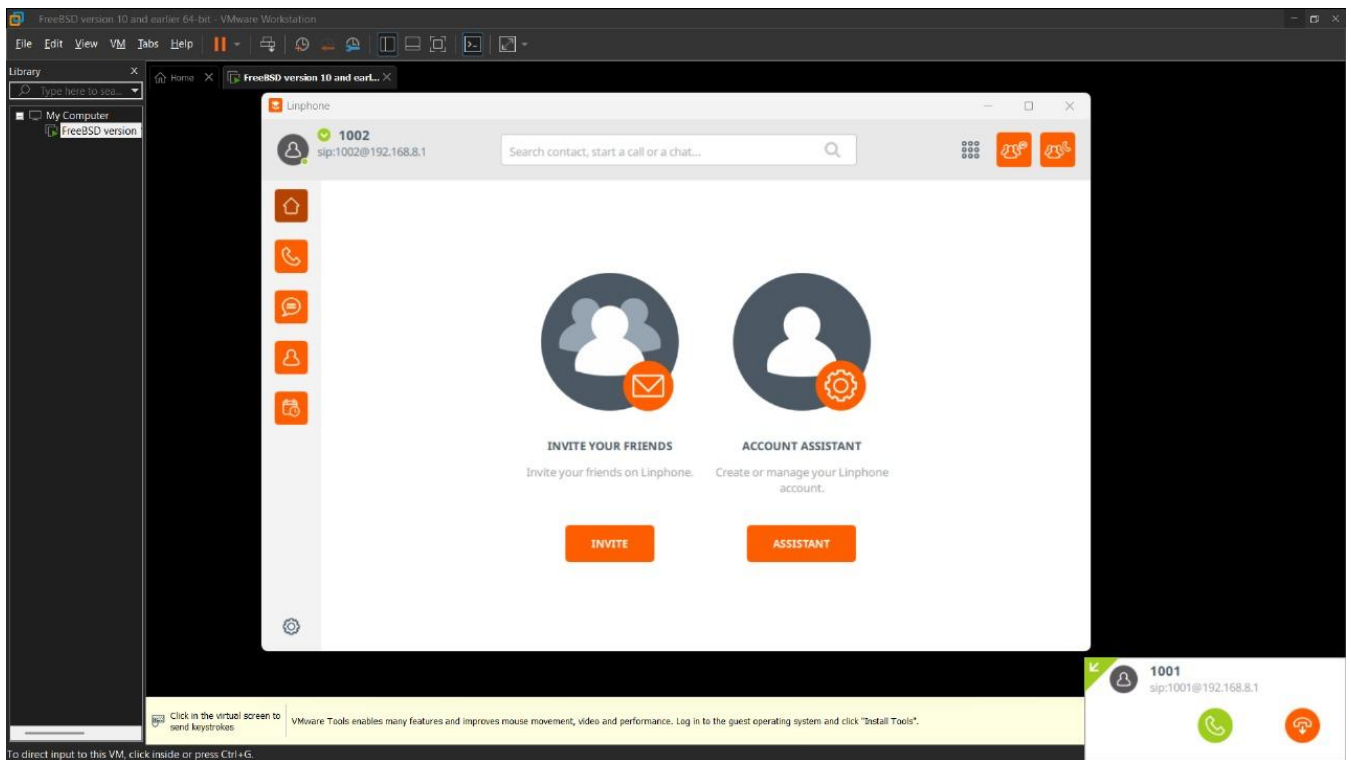
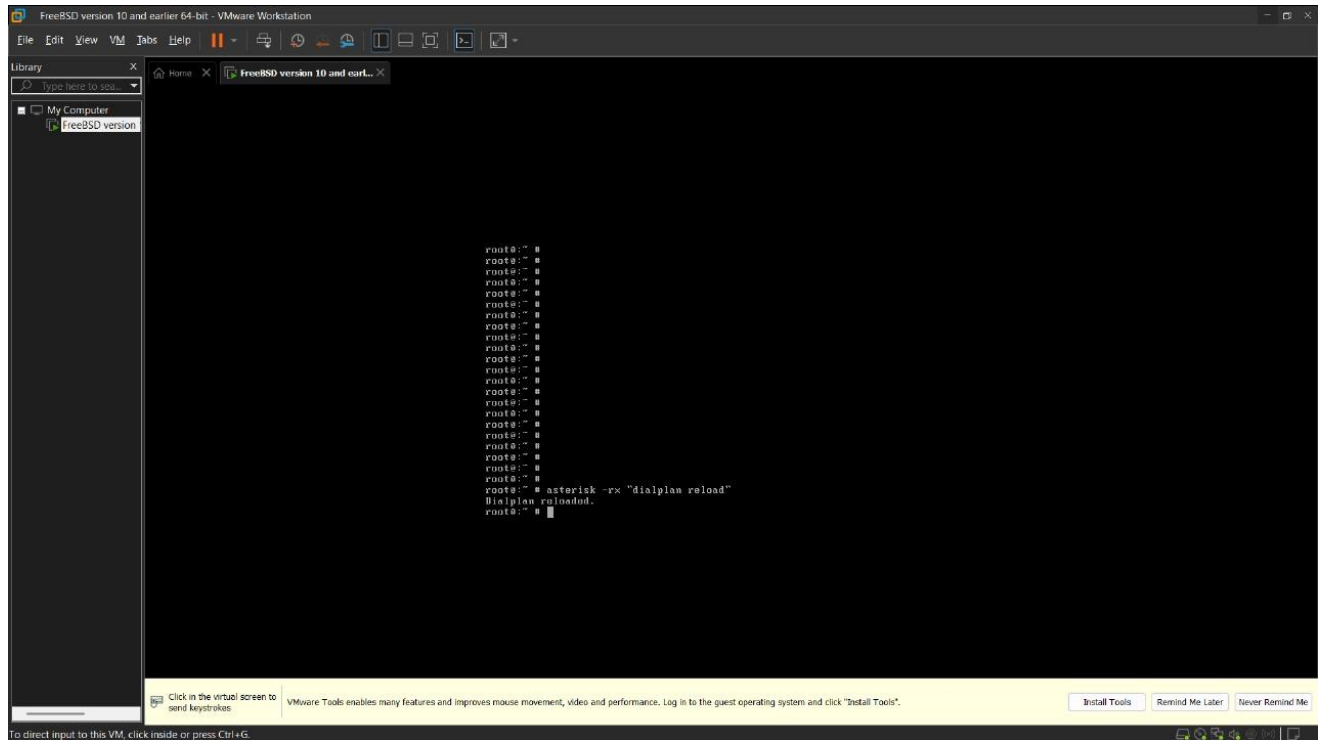
- 404 Not Found when 1002 not registered

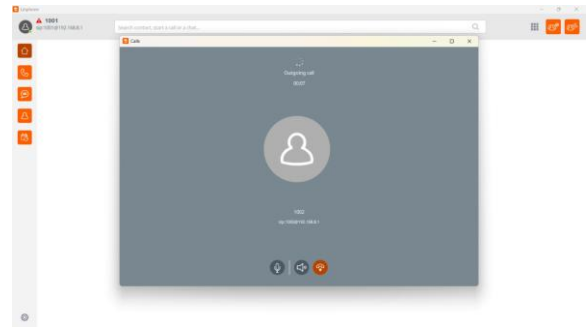
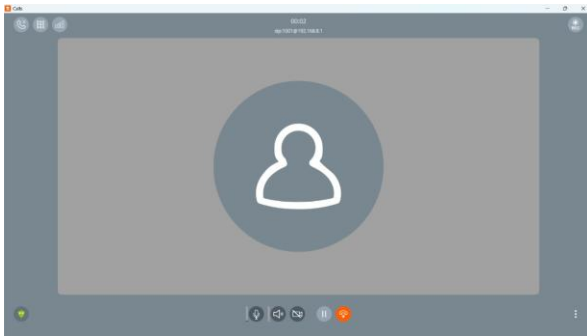
- After registration both sides, calls connected
```

Root cause analysis (errors):

- 404 Not Found arises when destination extension has no active contact (not registered).
- Fixes applied / Troubleshooting steps:
- Ensure both devices are registered (sip show peers shows address + OK).

- Use 'sip show peers' and 'sip show registry' to debug.





```
FreeBSD version 10 and earlier 64-bit - VMware Workstation
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*CLI> pjsip show transports
Transport: <TransportId.....> <Type> <cos> <tos> <BindAddress.....>
.....>
=====
Transport: transport-udp          udp          0          0 0.0.0.0:5060
Objects found: 1

*CLI> exit
asterisk cleanly ending (0).
Executing last minute cleanups
root@:~# asterisk -rx "pjsip reload"
Module 'res_pjsip.so' reloaded successfully.
Module 'res_pjsip_authenticator_digest.so' reloaded successfully.
Module 'res_pjsip_endpoint_identifier_ip.so' reloaded successfully.
Module 'res_pjsip_ua.so' reloaded successfully.
Module 'res_pjsip_notify.so' reloaded successfully.
Module 'res_pjsip_outbound_publish.so' reloaded successfully.
Module 'res_pjsip_publish_asterisk.so' reloaded successfully.
Module 'res_pjsip_outbound_registration.so' reloaded successfully.
root@:~#
```

Click in the virtual screen to send keystrokes. VMware Tools enables many features and improves mouse movement, video and performance. Log in to the guest operating system and click "Install Tools".

To direct input to this VM, click inside or press Ctrl+G.

```
Objects found: 2

*CLI> pjsip show transports

Transport: <TransportId.....> <Type> <cos> <tos> <BindAddress.....>
.....>
=====
Transport: transport-udp          udp          0          0 0.0.0.0:5060
Objects found: 1

*CLI> █
```

Lab 6 – PF Firewall Configuration

Objective: Use PF to restrict traffic to only necessary VoIP ports and management.

High-level steps:

- Edit /etc/pf.conf with whitelist rules for SIP and RTP.
- Enable pf and test connectivity.

Commands & configuration :

```
# Example /etc/pf.conf

# Block all by default

block in all

block out all

set skip on lo

# Allow SSH

pass in proto tcp from any to any port 22 keep state

# Allow SIP

pass in proto udp from any to any port 5060 keep state

# Allow RTP range

pass in proto udp from any to any port 10000:20000 keep state

# Allow DNS outbound

pass out proto { tcp udp } to any port 53 keep state
```

Expected output / behaviour:

```
# Expected pfctl -sr to list the rules and calls to work.

# netstat or sockstat should show asterisk bound to UDP 5060.
```

Observed output / errors:

```
# Observed

- After enabling PF, ping failed (ICMP blocked)
```

Root cause analysis (errors):

- ICMP is blocked by default when using a block all policy.
- Fixes applied / Troubleshooting steps:

- Add 'pass inet proto icmp all' to pf.conf and reload PF.
- Reload PF: service pf reload; verify rules: pfctl -sr

FreeBSD version 10 and earlier 64-bit - VMware Workstation

File Edit View VM Jobs Help

Library

Type here to search

My Computer

FreeBSD version

```

^_ (escape) ^mm ^y search prompt ^k delete line ^p prev li ^g prev page
^o ascii code ^x search ^i undelete line ^n next li ^v next page
^a end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete char ^l undelete char ESC-Enter: exit
=====line 1 col 0 lines from top 1 =====
# Block everything by default
block in all
block out all

# Allow localhost
set skip on lo

# Allow SSH
pass in proto tcp from any to any port 22 keep state

# Allow SIP
pass in proto udp from any to any port 5060 keep state
pass in proto tcp from any to any port 5060 keep state

# Allow TLS SIP
pass in proto tcp from any to any port 5061 keep state

# Allow RTP
file "/etc/pf.conf", 20 lines

```

To direct input to this VM, click inside or press Ctrl+G

[illegible]

Lab 7 – TLS for SIP (Attempt)

Objective: Configure TLS for SIP signaling using generated certs.

High-level steps:

- Generate CA and Asterisk server certificate using openssl.
- Configure sip.conf (chan_sip) or pjsip.conf (PJSIP) for TLS.
- Reload and test with TLS client (Zoiper TLS) on port 5061.

Commands & configuration :

```
# Generate CA and server cert

mkdir -p /usr/local/etc/asterisk/keys

cd /usr/local/etc/asterisk/keys

openssl genrsa -out ca.key 2048

openssl req -new -x509 -key ca.key -out ca.crt -days 365 -subj
"/C=US/ST=Test/L=Lab/O=MyPBX/CN=asterisk-ca"

openssl genrsa -out asterisk.key 2048

openssl req -new -key asterisk.key -out asterisk.csr -subj
"/C=US/ST=Test/L=Lab/O=MyPBX/CN=asterisk.local"

openssl x509 -req -in asterisk.csr -CA ca.crt -CAkey ca.key -CAcreateserial -
out asterisk.pem -days 365

# Example sip.conf TLS lines

[general]

tlsenable=yes

tlsbindaddr=0.0.0.0:5061

tlscertfile=/usr/local/etc/asterisk/keys/asterisk.pem

tlsprivatekey=/usr/local/etc/asterisk/keys/asterisk.key

tlscacfile=/usr/local/etc/asterisk/keys/ca.crt
```

Expected output / behaviour:

```
# Expected

sip show settings should list TLS cert paths; sockstat should show Asterisk
listening on :5061
```

Observed output / errors:

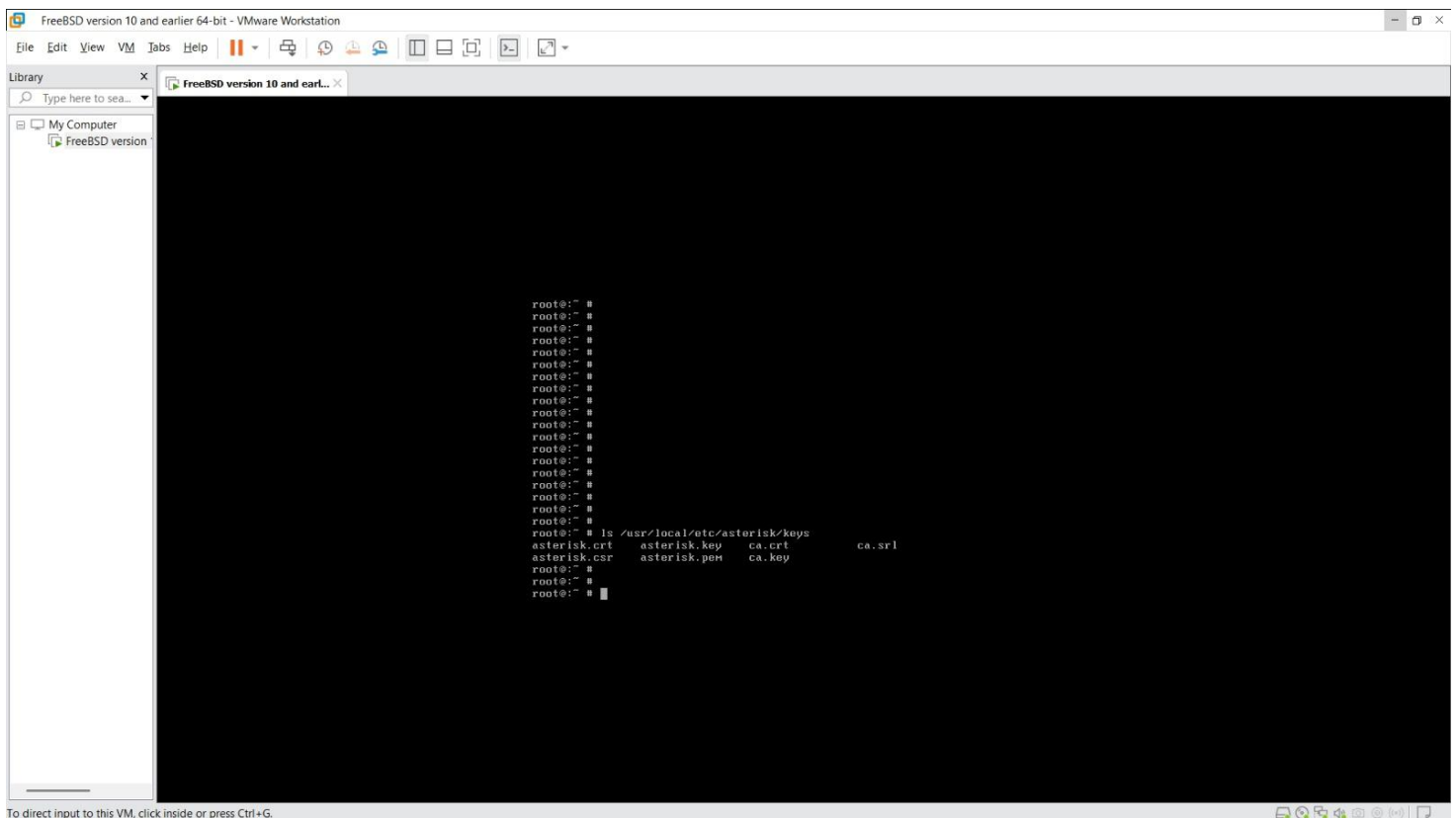
```
# Observed

- ldd /usr/local/lib/asterisk/modules/chan_sip.so -> no libssl/libcrypto
- pjsip/res_pjsip modules not present; 'pjsip show transports' not available
```

- sip show settings showed TLS bindaddr but no certfile/privatekey/ca entries

Root cause analysis (errors):

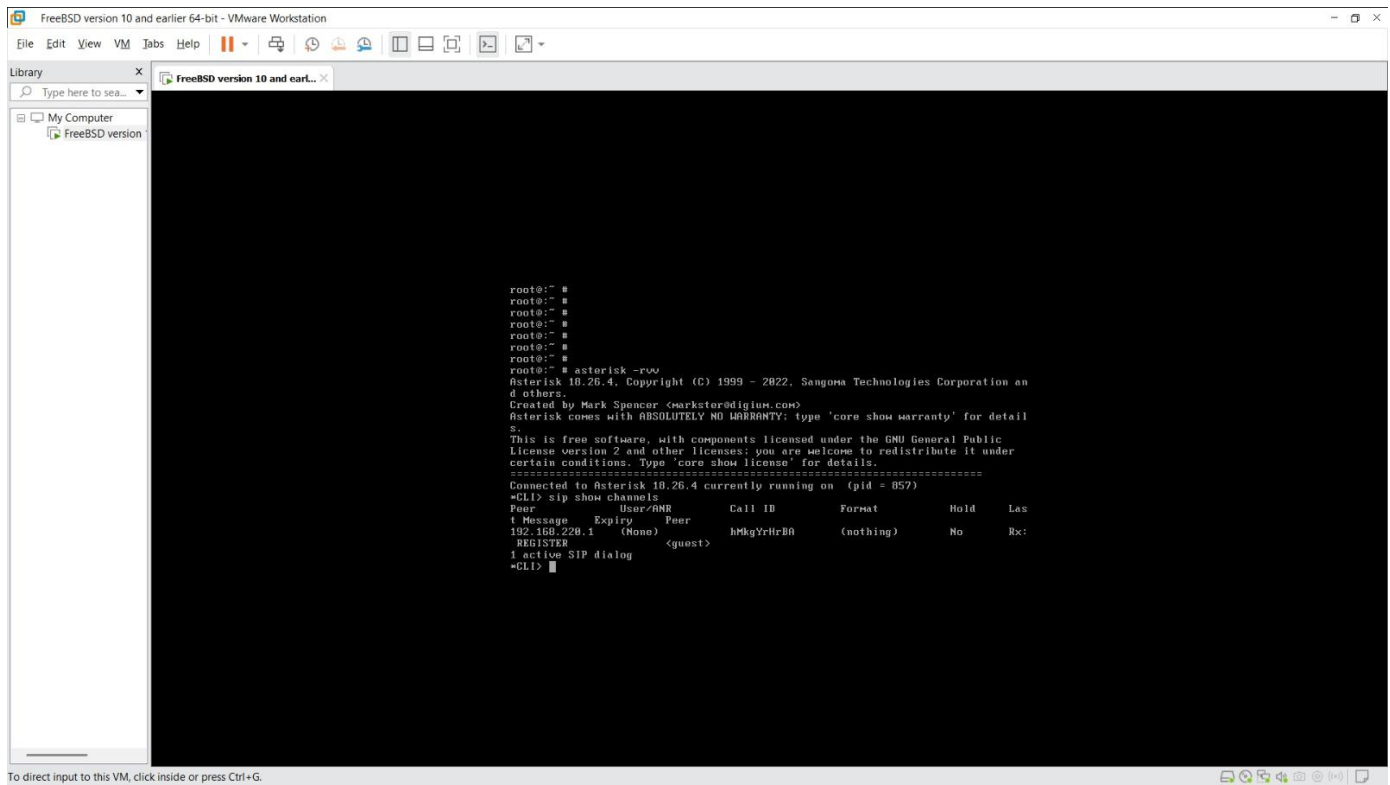
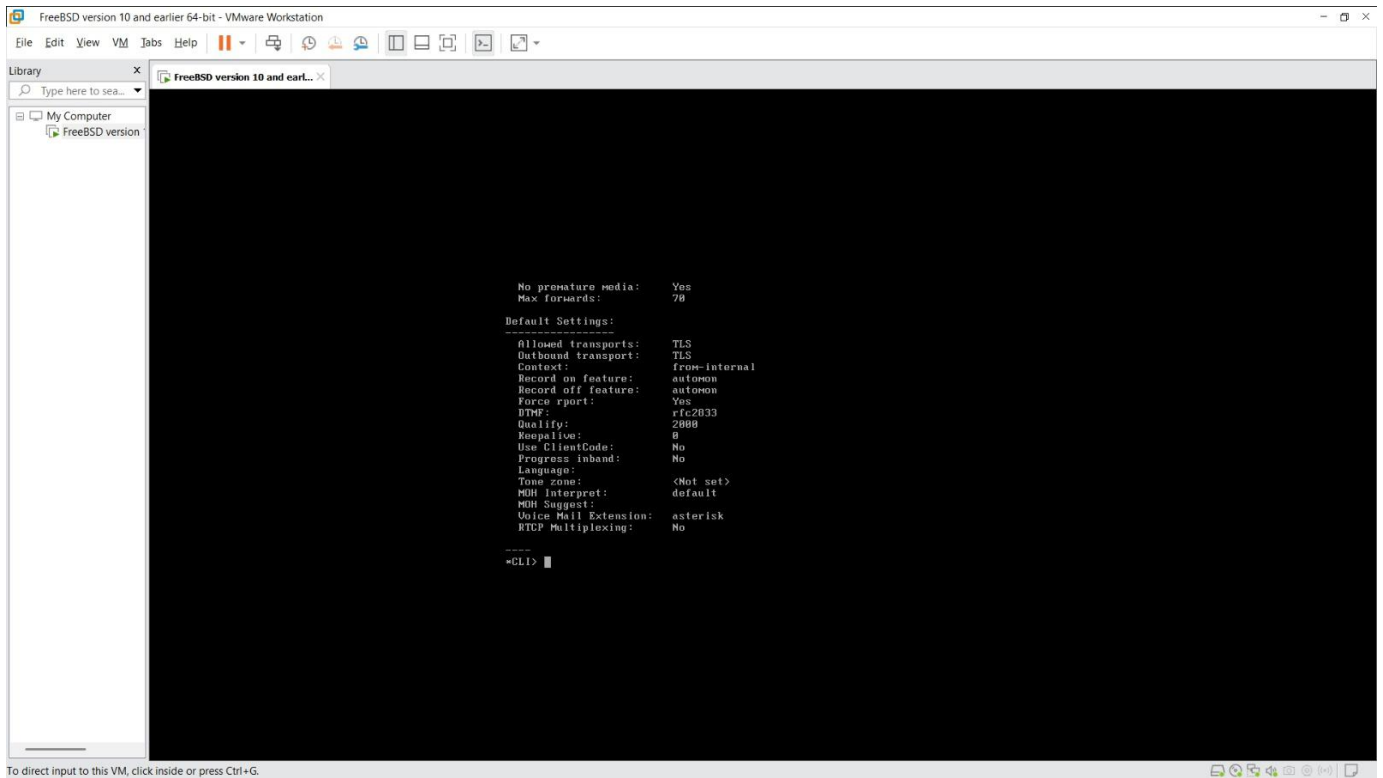
- Asterisk package was compiled without OpenSSL support for chan_sip; PJSIP wasn't installed in this package; therefore TLS cannot be used.
- Fixes applied / Troubleshooting steps:
- Two options: rebuild from ports enabling OPENSSL/PKCS11 and PJSIP, or document TLS configuration and use UDP fallback.
- Because of time and network constraints, proceeded with UDP and documented TLS config and missing support.



The screenshot shows a VMware Workstation window titled "FreeBSD version 10 and earlier 64-bit - VMware Workstation". The main area is a terminal window for a FreeBSD system. The prompt is "root@:~". The user has entered several commands, including "ls /usr/local/etc/asterisk/keys", which has produced the following output:

```
asterisk.crt  asterisk.key  ca.crt  
asterisk.csr  asterisk.pem  ca.key
```

The terminal window also shows a sidebar on the left with "My Computer" and "FreeBSD version" listed. At the bottom of the window, there is a status bar that reads "To direct input to this VM, click inside or press Ctrl+G."



FreeBSD version 10 and earlier 64-bit - VMware Workstation

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Library

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My Computer

FreeBSD version

```
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^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete char ^j undelete char ESC-Enter: exit
=====line 1 col 8 lines from top 1=====
[general]
context=from-internal
bindport=5805
bindaddr=0.0.0.0
encryption=yes
tlsenable=yes
tlshbindaddr=0.0.0.0:5861
tlscertfile=/usr/local/etc/asterisk/keys/asterisk.pem
tlsprikeyfile=/usr/local/etc/asterisk/keys/asterisk.key
tlscalfile=/usr/local/etc/asterisk/keys/ca.crt
tlscipher=RL
transport=tls
nat-force-external=yes
qualify=yes
[1001]
type=friend
host=dynamic
file=/usr/local/etc/asterisk/sip.conf", 40 lines
```

To direct input to this VM, click inside or press Ctrl+G.

FreeBSD version 10 and earlier 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to sea...

My Computer

FreeBSD version

```
^[(escape) menu ^y search prompt ^k delete line ^p prev li ^g prev page
^o ascii code ^x search ^l undelete line ^u next li ^v next page
^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete char ^j undelete char ESC-Enter: exit
=====line 33 col 21 lines from top 33=====
[1001]
type=friend
host=dynamic
secret=pass1001
context=from-internal
disallow=all
allow=ulaw
transport=tls
encryption=yes
avpf=yes
icesupport=yes
media_encryption=sdes
[1002]
type=friend
host=dynamic
secret=pass1002
context=from-internal
```

To direct input to this VM, click inside or press Ctrl+G.

```

FreeBSD version 10 and earlier 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
x
FreeBSD version 10 and ear...
My Computer
FreeBSD version

root@:~ # mkdir -p /usr/local/etc/asterisk/keys
root@:~ # cd /usr/local/etc/asterisk/keys
root@:/usr/local/etc/asterisk/keys # openssl genrsa -out ca.key 2048
Generating RSA private key, 2048 bit long modulus (2 primes)
.....+++++
e is 65537 (hex10001)
root@:/usr/local/etc/asterisk/keys # openssl req -new -x509 -key ca.key -out ca.
crt -days 365 -subj "/C=US/ST=Test/L=Lab/O=MyPBX/CN=asterisk-ca"
req: Use -help for summary.
root@:/usr/local/etc/asterisk/keys # openssl genrsa -out asterisk.key 2048
Generating RSA private key, 2048 bit long modulus (2 primes)
.....+++++
e is 65537 (hex10001)
root@:/usr/local/etc/asterisk/keys # openssl req -new -key asterisk.key -out ast
erisk.csr -subj "/C=US/ST=Test/L=Lab/O=MyPBX/CN=192.168.220.120"
req: Use -help for summary.
root@:/usr/local/etc/asterisk/keys # openssl x509 -req -in asterisk.csr -CA ca.c
rt -CAkey ca.key -CAcreateserial -out asterisk.crt -days 365
x509: Unknown parameter -out
x509: Use -help for summary.
root@:/usr/local/etc/asterisk/keys # cat asterisk.key asterisk.crt > asterisk.pe
M
cat: asterisk.crt: No such file or directory

```

Click in the virtual screen to send keystrokes. VMware Tools enables many features and improves mouse movement, video and performance. Log in to the guest operating system and click "Install Tools".

To direct input to this VM, click inside or press Ctrl+G.

Lab 8 – SRTP (Media Encryption)

Objective: Configure SRTP so that RTP streams are encrypted (if SRTP module present).

High-level steps:

- Enable encryption options in sip.conf or pjsip.conf.
- Configure endpoints to prefer SRTP.
- Test call and check RTP Encryption indicators in sip show channels.

Commands & configuration (copy-paste):

```

# Example chan_sip settings

[general]

encryption=no ; default off if res_srtp not available

[1001]

encryption=yes

avpf=yes

```

```
media_encryption=sdes
```

Expected output / behaviour:

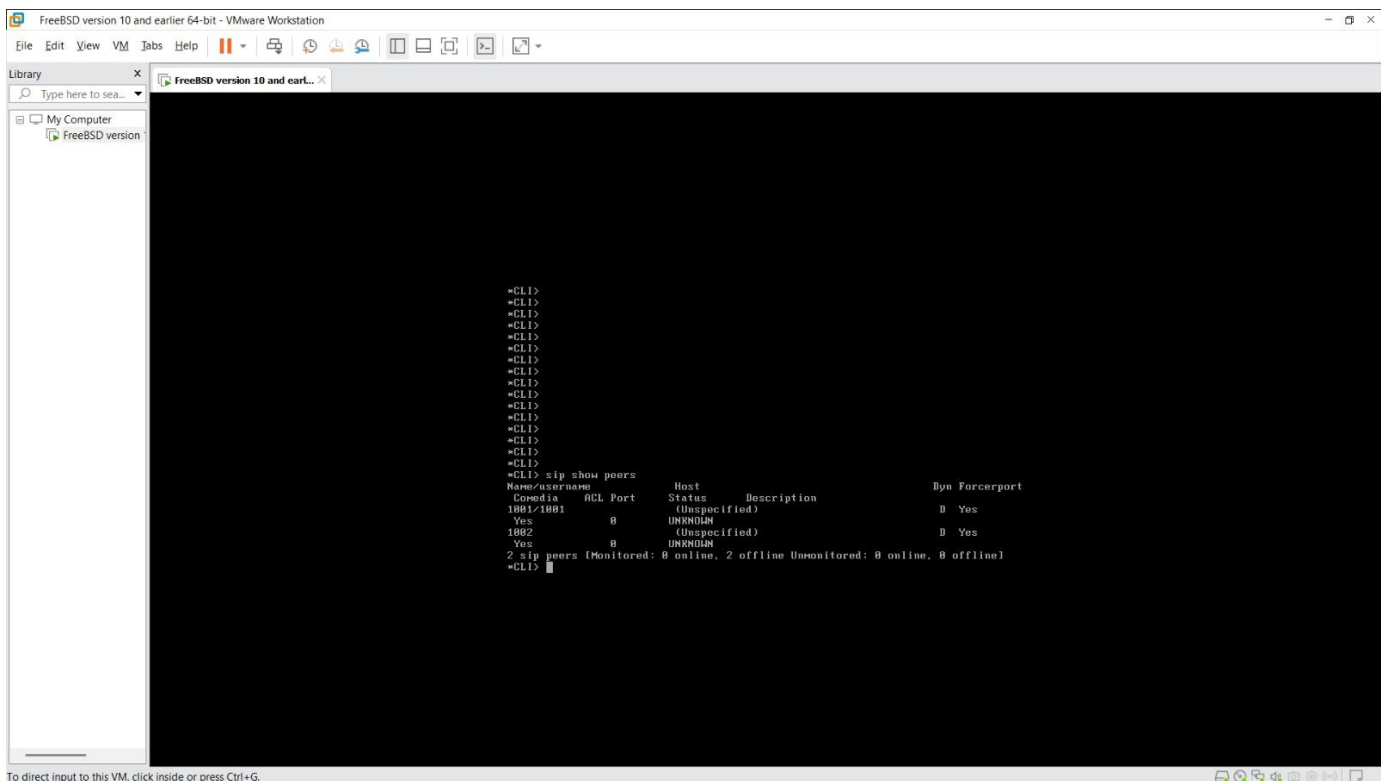
```
# Expected: res_srtp.so loaded and calls show RTP Encryption: Yes
```

Observed output / errors:

```
# Observed: res_srtp.so missing; RTP unencrypted
```

Root cause analysis (errors):

- SRTP module (res_srtp) not present in the pkg build; requires building Asterisk with SRTP/PJPROJECT support.
- Fixes applied / Troubleshooting steps:
- Documented SRTP config; to enable, rebuild Asterisk with SRTP support or install a package variant that includes res_srtp.



Lab 9 – NAT & Security Best Practices

Objective: Harden Asterisk, configure NAT traversal, and protect against brute force.

High-level steps:

- Configure nat and qualify in sip.conf.
- Configure VMware NAT port forwarding if using NAT.

- Install and configure Fail2ban to block repeated SIP login failures.

Commands & configuration (copy-paste):

```
# sip.conf NAT options

[general]

nat=force_rport,comedia

qualify=yes

# Example Fail2ban jail.local snippet

[asterisk-iptables]

enabled = true

filter = asterisk

action = iptables-allports[name=ASTERISK, protocol=all]

logpath = /var/log/asterisk/messages

maxretry = 5
```

Expected output / behaviour:

```
# Expected: Calls work across NAT; Fail2ban bans IP after repeated failed auths
```

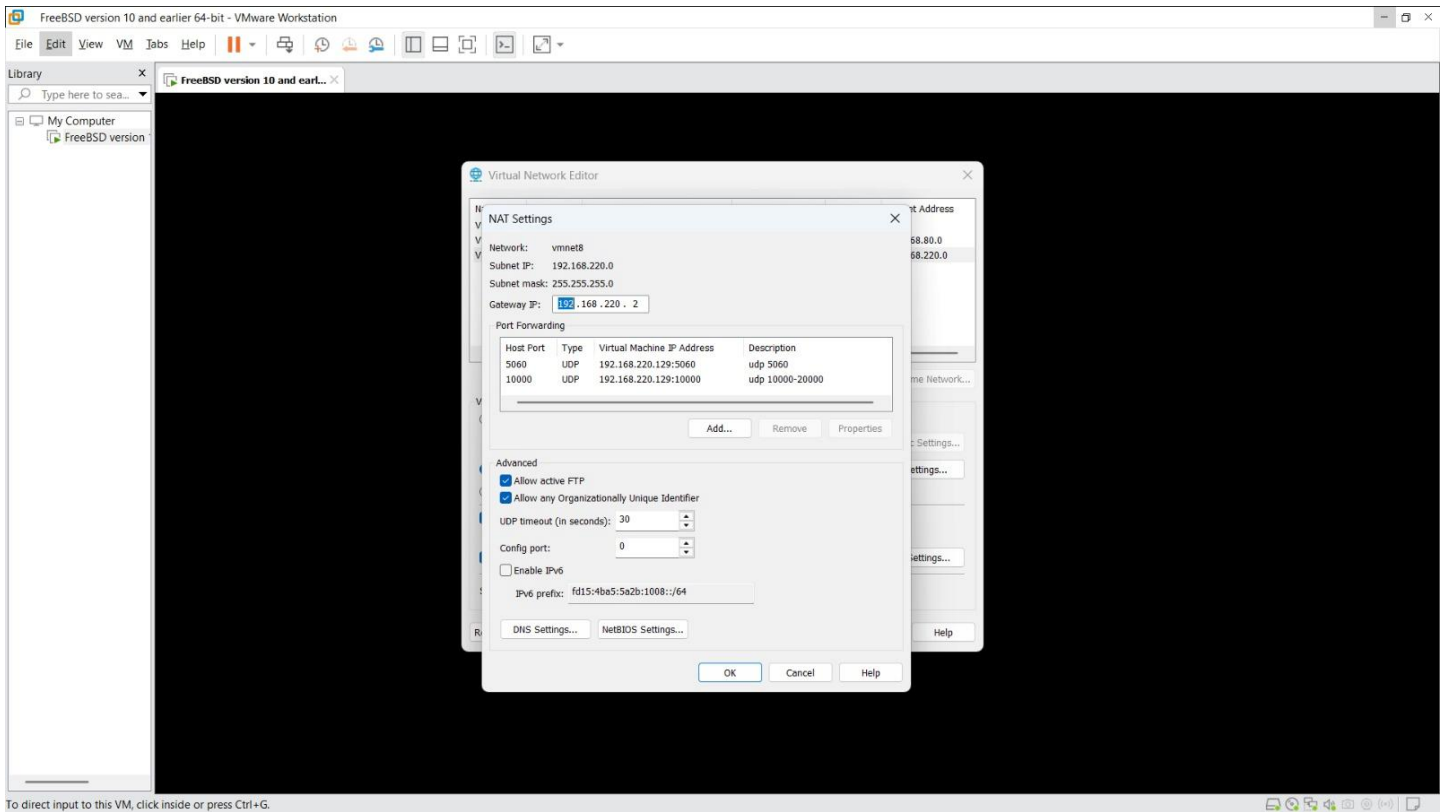
Observed output / errors:

```
# Observed: NAT forwarding was complex in VMware; bridged mode simpler.
Fail2ban logs showed blocked IPs after configuration.
```

Root cause analysis (errors):

- Port forwarding rules in VMware can be error-prone. Bridged mode is simpler for local testing with mobile clients.
- Fixes applied / Troubleshooting steps:
- If NAT must be used, ensure host-to-guest forwarding is configured for UDP 5060 and RTP range (10000-20000).

- Configure Fail2ban and check jail status: `service fail2ban status; tail -f /var/log/fail2ban.log`



Appendix A Error Log (Detailed)

Error	Symptom	Root Cause	Fix	Related Lab
408 Request Timeout	REGISTER sent but no reply	NAT/bridged mismatch or firewall blocking	Switch to Bridged or open port forwarding for UDP 5060	See above
401 Unauthorized	Authentication failed	Wrong secret or auth config mismatch	Correct secret in sip.conf and Zoiper	See above
404 Not Found	Dial to unregistered extension	Destination not registered	Register destination, verify sip show peers	See above

No route to host	Ping to external IP failed	Missing default gateway	Add route add default <gateway> or enable DHCP default route	See above
ICMP blocked by PF	ping blocked after enabling PF	PF block policy	Add 'pass inet proto icmp all' to /etc/pf.conf	See above
TLS not available	sip show settings no cert paths; ldd chan_sip.so no libssl	pkg build without OpenSSL	Rebuild from ports enabling OPENSLL or use a distro/package with TLS	See above
res_pjsip missing	pjsip show transports not available	Asterisk package lacks PJSIP	Reinstall from ports with PJPROJECT/PJSIP enabled	See above
res_srtp missing	No SRTP module	Pkg build lacks SRTP	Rebuild with SRTP or install appropriate package	See above

Appendix B — Configuration Files (copy-paste ready)

1) /usr/local/etc/asterisk/sip.conf (chan_sip example)

```
[general]

context=from-internal

bindport=5060

bindaddr=0.0.0.0

transport=udp

tlsenable=yes

tlsbindaddr=0.0.0.0:5061

tlscertfile=/usr/local/etc/asterisk/keys/asterisk.pem

tlsprivatekey=/usr/local/etc/asterisk/keys/asterisk.key

tlscacfile=/usr/local/etc/asterisk/keys/ca.crt
```

```
tlscipher=ALL

[1001]

type=friend

host=dynamic

secret=pass1001

context=from-internal

disallow=all

allow=ulaw

[1002]

type=friend

host=dynamic

secret=pass1002

context=from-internal

disallow=all

allow=ulaw
```

2) /usr/local/etc/asterisk/extensions.conf (dialplan)

```
[from-internal]

exten => 100,1,Answer()

    same => n,Playback(hello-world)

    same => n,Hangup()

exten => 1001,1,Dial(SIP/1001,30)

    same => n,Hangup()

exten => 1002,1,Dial(SIP/1002,30)

    same => n,Hangup()
```

3) /etc/pf.conf (example PF rules)

```
block in all

block out all
```

```

set skip on lo

# Allow SSH

pass in proto tcp from any to any port 22 keep state

# Allow SIP (UDP 5060)

pass in proto udp from any to any port 5060 keep state
pass in proto tcp from any to any port 5060 keep state

# Allow RTP media

pass in proto udp from any to any port 10000:20000 keep state

# Allow DNS outbound

pass out proto { tcp udp } to any port 53 keep state

# Allow ICMP for troubleshooting

pass inet proto icmp all

```

4) Fail2ban jail.local example

```

[asterisk-iptables]

enabled = true

filter = asterisk

action = iptables-allports[name=ASTERISK, protocol=all]

logpath = /var/log/asterisk/messages

maxretry = 5

```

Appendix C — VMware NAT Port Forwarding (example)

If using VMware NAT (not bridged), configure NAT port forwarding on the host to reach the guest:

Example rules (host -> guest 192.168.220.129):

```

# UDP 5060 (SIP signaling)

HostPort 5060 -> GuestPort 5060 (UDP)

# RTP range 10000-20000

HostPort 10000-20000 -> GuestPort 10000-20000 (UDP)

```

Note: Steps differ by VMware product. In VMware Workstation player: VM > Settings > Network Adapter > NAT > NAT Settings > Port Forwarding.

Final Notes & Submission Checklist

Checklist:

- - Insert actual screenshots into the placeholders.
- - Add screen recordings (link or embedded) showing: Zoiper registration, call, PF reload + call, Fail2ban blocking.
- - Run the verification commands listed in each lab and include their outputs as screenshots.
- - If you rebuild Asterisk for TLS/SRTP, replace the "Observed" fields in Lab 7/8 with the new outputs.

Video Link : <https://drive.google.com/drive/folders/10SAwBeUM1npFHJ-BMIPwsJ8KBc16-rN5?usp=sharing>

References :- <https://chatgpt.com/share/68b1780f-0094-8008-921f-381105e432f9>

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End of report.